

Matrices with Squarefree Characteristic Polynomial: A Disproof of Conjecture 00000001854

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Abstract

The source asserts that the number of $n \times n$ matrices over $\mathbb{F}_q$ with squarefree characteristic polynomial is exactly $q^{n^2} \prod (1 - q^{-i^2})$. For the product over $1 \leq i \leq n$ we show that the identity fails for every finite field and every $n \geq 1$: for $n = 1$ the count is q and the formula gives $q - 1$, and for $n \geq 2$ the formula is not an integer. At $n = 1$ the identity also fails for every non-empty truncation of the product and for the infinite product, and the product over $1 \leq i \leq n - 1$ fails at $n = 5$. These statements are verified in Lean with Mathlib's matrices, characteristic polynomials and squarefree elements.

Notation and reading of the source. For a finite field F with q elements and $n \geq 0$ let

$$N_n(F) = \#\{M \in \text{Mat}_n(F) : \chi_M \text{ is squarefree}\},$$

where $\chi_M = \det(XI - M) \in F[X]$ and a polynomial is squarefree if no square of a non-unit divides it. Both language versions of the source state

$$N_n(\mathbb{F}_q) = q^{n^2} \prod (1 - q^{-i^2})$$

as an exact closed form, without giving the range of the index i . The shape of the formula imitates $|\text{GL}_n(\mathbb{F}_q)| = q^{n^2} \prod_{i=1}^n (1 - q^{-i})$, so the natural reading is the product over $1 \leq i \leq n$ (Reading A). The other reading we treat is the infinite product over $i \geq 1$ (Reading B). The claim is read as an identity for all prime powers q and all $n \geq 1$.

Theorem 1. *For every finite field F with q elements, $N_1(F) = q$.*

Proof. The characteristic polynomial of the 1×1 matrix (a) is $X - a$. A polynomial of degree one over a field is irreducible, and an irreducible element is squarefree: if d^2 divides $X - a$ then d divides an irreducible element twice, which forces d to be a unit. Hence all q matrices are counted. $\square$

Theorem 2 (Reading A). *Let F be a finite field with q elements and let $n \geq 1$. Then*

$$N_n(F) \neq q^{n^2} \prod_{i=1}^n (1 - q^{-i^2}).$$

For $n = 1$ the right-hand side is $q - 1$, and for $n \geq 2$ it is not an integer.

Proof. For $n = 1$ the right-hand side is $q(1 - q^{-1}) = q - 1$, while $N_1(F) = q$ by Theorem 1.

Let $n \geq 2$ and put $S = \sum_{i=1}^n i^2$. Suppose that the right-hand side equals an integer N . Multiplying by $q^S = \prod_{i=1}^n q^{i^2}$ gives the identity of integers

$$N q^S = q^{n^2} \prod_{i=1}^n (q^{i^2} - 1).$$

Since $n \geq 2$, the sum S contains the two different terms 1^2 and n^2 , so $S \geq n^2 + 1$. Cancelling q^{n^2} shows that q divides $P = \prod_{i=1}^n (q^{i^2} - 1)$. But every factor is coprime to q , because $q^{i^2-1} \cdot q - (q^{i^2} - 1) = 1$ for $i \geq 1$; hence P is coprime to q . A common divisor q of q and P must then be a unit, contradicting $q \geq 2$. $\square$

Theorem 3 (Other finite ranges). *Let $q \geq 2$ and let $n, m \geq 1$ with $\sum_{i=1}^m i^2 > n^2$. Then $q^{n^2} \prod_{i=1}^m (1 - q^{-i^2})$ is not an integer. In particular, with the product over $1 \leq i \leq n-1$ the identity fails at $n = 5$ for every finite field, because $1 + 4 + 9 + 16 = 30 > 25$.*

Proof. The proof of Theorem 2 for $n \geq 2$ used only $S > n^2$, and it applies with $S = \sum_{i=1}^m i^2$ and the product over $1 \leq i \leq m$. $\square$

Theorem 4 (Truncations and Reading B at $n = 1$). *Let F be a finite field with q elements.*

1. *For every $m \geq 1$, $q \prod_{i=1}^m (1 - q^{-i^2}) \leq q - 1 < q = N_1(F)$.*
2. *If the partial products $\prod_{i=1}^m (1 - q^{-i^2})$ converge to L as $m \rightarrow \infty$, then $qL \leq q - 1 < N_1(F)$.*

Proof. Every factor $1 - q^{-i^2}$ lies in $[0, 1]$, so the partial products are non-negative and non-increasing in m . For $m \geq 1$ they are therefore at most the first factor $1 - q^{-1}$, which gives (1). The limit of a sequence that is eventually at most $1 - q^{-1}$ is at most $1 - q^{-1}$, which gives (2). $\square$

Thus the asserted identity is false in Reading A for every q and every $n \geq 1$, and in Reading B at $n = 1$ for every q .

Remark 1 (The true values for small n ; not formalised). For $n = 2$ one has $N_2(\mathbb{F}_q) = q^4 - q^3$. Indeed, a monic quadratic is not squarefree exactly when it equals $(X - a)^2$ for some $a \in \mathbb{F}_q$, and a is then unique. A matrix M has $\chi_M = (X - a)^2$ exactly when $M - aI = \begin{pmatrix} x & y \\ z & -x \end{pmatrix}$ with $x^2 + yz = 0$. This equation has q^2 solutions: $q(q - 1)$ with $y \neq 0$ (then $z = -x^2/y$) and q with $y = 0$ (then $x = 0$). So $q \cdot q^2$ matrices are excluded. Reading A gives $(q^4 - q^3)(1 - q^{-4})$ instead. The supplementary script `verify.py` enumerates all matrices for small parameters and finds $N_1 = q$ and $N_2 = q^4 - q^3$ for $q \in \{2, 3, 5\}$, $N_3(\mathbb{F}_2) = 160$ and $N_3(\mathbb{F}_3) = 11178$. In particular a product over $1 \leq i \leq n - 1$, which happens to agree with N_n for $n \leq 2$, gives $240 \neq 160$ at $(n, q) = (3, 2)$. This remark is supported by the paper argument and the enumeration only; it is not used in the theorems above.

The earlier submission and its error

Pull request #23 (orionsheep) treated this conjecture with the same counterexample at $n = 1$ and the same two readings. It was merged on 2026-10-01 and removed by the organisers in commit 4504549 the same day; the stated reason was that its Lean theorems carried none of the mathematical objects. The mathematics of that submission was correct. The error was in its Lean file, which used Lean core only. The squarefree predicate was defined as the constant function

```
def sqfree1b : Fin 2 -> Bool := fun _ => true,
```

the count was the length of a two-element list filtered by this predicate, and the decisive statements were closed arithmetic facts such as $(2 : \text{Nat}) \neq 1$. No matrix, characteristic polynomial, squarefreeness, finite field or quantifier over q and n occurred. The present submission defines the count with Mathlib's `Matrix.charpoly` and `Squarefree`, proves $N_1(F) = q$ for every finite field, and proves the failure of the identity for all finite fields and all $n \geq 1$.

Formal verification and scope

The Lean definition

```
sqfreeCount F n
```

is, for an arbitrary field F , the cardinality of the subtype of

`Matrix (Fin n) (Fin n) F`

on which `Squarefree M.charpoly` holds. The right-hand side is `formula q n m`, the rational number $q^{n^2} \prod_{i=1}^m (1 - 1/q^{i^2})$, and `partialProduct q m` is the real partial product.

The theorems

`squarefree_charpoly_one, sqfreeCount_one`

are Theorem 1 for every finite field. The theorem

`formula_not_natural`

states that for $q \geq 2$ and $n \geq 2$ no natural number equals `formula q n n`. The theorem `formula_fails` is Theorem 2: for every type `F` with `Field` and `Fintype` instances and every $n \geq 1$,

`(sqfreeCount F n : Rat) ≠ formula (Fintype.card F) n n.`

Theorem 3 is

`formula_not_natural_of_sum, shifted_formula_fails,`

the second being the case $n = 5, m = 4$ for every finite field. Theorem 4(1) is

`truncated_formula_fails_at_one,`

and Theorem 4(2) is

`infinite_product_fails_at_one,`

with convergence expressed by `Filter.Tendsto`. The propositions

`ClaimedClosedForm, ClaimedClosedFormInfinite`

are the universally quantified identities of Readings A and B. Their negations are the final theorems

`conjecture_false, conjecture_false_infinite.`

The negations need one concrete finite field. The `Mathlib` module that declares the field structure on `ZMod p` is not among the modules imported here, so the file declares the instance `instFieldZModTwo` from `Mathlib`'s commutative ring structure and inversion on `ZMod 2`, proving $aa^{-1} = 1$ for $a \neq 0$. The general theorems do not depend on this instance. No `native_decide`, `sorry` or custom axiom occurs.

Scope. The source does not fix the index range of the product. The formal results cover the product over $1 \leq i \leq n$ for all $n \geq 1$, and at $n = 1$ every truncation $1 \leq i \leq m$ with $m \geq 1$ and the infinite product. Reading B is refuted only at $n = 1$, which suffices for a claim about all n . The product over $1 \leq i \leq n - 1$, which is empty at $n = 1$ and agrees with N_n for $n \leq 2$, is refuted formally at $n = 5$ by Theorem 3. More generally a range $1 \leq i \leq m(n)$ is refuted as soon as $m(1) \geq 1$, or $\sum_{i \leq m(n)} i^2 > n^2$ for some n . Index ranges that do not start at $i = 1$ are not covered. No claim is made about the asymptotic proportion of such matrices as n or q grows.

Source and reproduction

The exact bilingual source is included byte-for-byte as `SOURCE.md`. Its repository path is

`conjectures/00000001854.md.`

The project pins Lean 4.19.0 and `Mathlib` commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b.`

From `lean/`, run `lake build`, then

`lake env lean -DwarningAsError=true Main.lean.`

The included public-Git manifest locks all dependency revisions. The `verification/` directory records the fresh-build logs, theorem-axiom reports, artifact hashes and the self-review. The script `verify.py` is supplementary and supports only the remark.